



CHAPTER

16 The Sources of Memory Errors

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Abstract

This chapter discusses some of the various types of errors that are the by-products of highly efficient and practical human memory systems. The errors vary in size, commonness, and potential for impact. Errors may be of omission (forgetting details and whole events that did happen), commission (remembering details and whole events that did not actually happen), or a combination of the two. Errors can happen at any stage from initial perception to final memory retrieval. Critically, some errors are so compelling for the rememberer that they cannot be “undone” once they have occurred.

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Several of the chapters in this book are a testament to the complexities of human memory, as well as to its power to hold tremendous amounts of information for impressive periods of time. This chapter will discuss some of the downsides of this complexity and capacity—that is, the errors that can result. In particular, underlying commonalities in apparently different errors will be noted and discussed.

Memory errors include both errors of omission and of commission. Errors of omission that will be discussed here include failures to initially encode information and failures to correctly retrieve it later. Errors of commission to be discussed include the adoption of incorrect details into existing memories and the creation of completely new, wholly false, memories.

Memory errors are part of daily living. Some of these errors are small and correctable, like temporarily forgetting a particular word or name. Errors that are not corrected may go completely unnoticed. Other errors are substantial and can have huge impacts on people’s lives, even including convictions for crimes committed by someone else. Part of the reason that these latter memory errors can have such consequences is that they can be extremely compelling for those who have them (and those who hear about them). Some of the factors that can make false memories so compelling will be discussed later in this chapter.

Contrary to common intuition, forgetting is a very important part of remembering. As William James (1890) put it, “If we remembered everything, we should on most occasions be as ill off as if we remembered nothing” (p. 68). If this seems strange, consider briefly what would happen if every memory of parking your car was as vivid as every other memory of parking your car, and then try to consider how you might go about finding your way home.

Instead of remembering with equal strength everything that has ever happened to us, and everything we have ever learned, we have a necessarily incomplete record of the past, plus a set of resources to help us organize and make sense of these incomplete records. These resources include schemata, ↪ biases of

various kinds, abilities to prioritize, assumptions, and reconstructive capabilities. These resources are mostly highly functional and make practical memory possible. But they also backfire in predictable ways and thus create predictable and testable patterns of errors (see also Schacter, 2001, for a further discussion of the adaptiveness of memory errors).

Because memory is a series of processes—broadly encoding, storage, and retrieval—there are a variety of opportunities for errors. In order for a memory to be successfully formed, saved, and accessed when needed, a number of things have to go right. Notably, the information or event has to be perceived (and thus to be readily perceptible), attended to, stored (and perhaps rehearsed) until needed, found again when required, and brought back into awareness. Each of these steps introduces error potential, and although these errors work somewhat differently from one another, they also share a variety of often telling commonalities.

As we will see later, errors in human memory are particularly likely in the presence of several factors. Memory errors are more likely when contradictory evidence is lacking (often because it was not recorded or has since been lost), when erroneous information fits in with our understanding of the world and expectations about it, when we are motivated to believe the erroneous information, and when the erroneous information comes from reliable sources. This is not to say that all of these factors must be present for a memory error to occur, but rather that the presence of more factors makes errors (and more substantial errors) more likely.

Incomplete Records

At any given moment in time, there are thousands if not millions of bits of information in one's immediate surroundings that one might attend to. Even in the relatively calm and banal environment of an academic office, for example, there are normally sights (computers, books, lights, pens, papers, knick-knacks), sounds (computer fans and keystrokes, vents, office neighbors), and smells (dust, caffeinated beverages) that one might notice. Taking in all of this sensory information continually in a meaningful way is simply not possible. And even attempting to do so would mean that we had little time for anything else. So we do not constantly attend to all of this input. Instead, we pay attention to what matters in the moment. But because only parts of the available information are attended to, there is a necessarily incomplete record available later. Schacter (1999, 2001) calls this the memory “sin” of absent-mindedness.

More precisely, sensory memory holds a huge amount of rich information but only for very short periods of time (see Cowan, 2008). From here, information that is attended to is processed by working memory and is thus more likely to end up in longer term store (Mulligan, 2008; Schacter, 2001), while information that is not attended to may well be lost forever (see Cowan, 2008). The particular role of attention in perceptual processes has been a major source of debate in perception research for decades (e.g., Mack & Rock, 1998; Rensink, 2002).

There are a variety of factors that affect the extent to which we pay attention to information, and whether this information is subsequently stored in memory. The first such factor is the simple availability of attention. Dividing attention between multiple tasks limits the amount of information that can be held on to for the long term (e.g., Craik, Govoni, Naveh-Benjamin, & Anderson, 1996). (This is something that frequent multitaskers might want to keep in mind.)

For example, Fernandes, Craik, Bialystok, and Kreuger (2007) gave subjects lists of words to learn and then repeat. Each list of 20 words was presented aurally and all the words in each list were from a single category (e.g., animals, tools). For some word lists, subjects were asked to make judgments about other, visually presented words (e.g., Is the object this word represents smaller or larger than a computer mouse?) at the same time as they listened to the word lists—that is, they were asked to encode the word lists under divided attention. Relative to conditions in which they were able to give the word lists their full attention, subjects were able to remember fewer words that they had learned in this divided attention condition. Divided attention has also been shown to influence retrieval, though not as strongly (Baddeley, Lewis, Eldridge, & Thomas, 1984).

In cases where sufficient attention is available, how that attention is utilized can make a substantial difference. Craik and Lockhart (1972) demonstrated that attending to shallow dimensions of stimuli, for example, whether to-be-remembered words are in *CAPITAL LETTERS* or *lowercase letters*, leads to relatively poor memory for those stimuli later. By contrast, deeper processing of the same information, for example

by semantically categorizing words, leads to better memory. This helps to explain why meaningful events are typically more memorable than nonmeaningful events—they are processed more deeply (e.g., Elias & Perfetti, 1973). Deep processing of information apparently leads to better memory because it facilitates the making of connections between new and old information (see Reisberg, 2010).

Further evidence that attention can influence what is originally encoded comes from recent research on the memory implications of change blindness and inattention blindness (see Laney & Loftus, 2010). A person is change blind when he or she fails to notice a significant change in the visual environment. For example, Levin, Simons, Angelone, and Chabris (2002) brought undergraduate volunteers into a lab for an experiment. An experimenter briefly explained the study to each participant from behind a counter, and then ducked below the counter on the pretence of picking up a stack of study materials. During this very brief interlude, the experimenter changed places with another individual, who had been hiding behind the counter out of view of the subject. This second individual popped up with the study materials and continued the conversation with the subject, as if nothing unusual had happened. Fifteen of the 20 subjects in the study failed to notice the change of experimenters, even though they were in the middle of a face-to-face conversation when the change occurred.

Inattention blindness is a similar phenomenon, whereby a person fails to notice an event entirely. The classic experimental demonstration of this involves a woman in a gorilla suit who subjects fail to notice in a video—even though she walks right into the middle of the scene and waves—because they are attending to people dressed in white and intentionally ignoring people dressed in black (Simons & Chabris, 1999).

It is also worth noting that there is a related line of research showing that people are particularly bad at predicting the errors that have been demonstrated in change blindness and inattention blindness research. This “change blindness blindness” work shows that people fail to predict that they would be change blind in scenarios where change blindness is, in fact, quite likely (Beck, Levin, & Angelone, 2007).

The memory implications of change blindness and inattention blindness are perhaps obvious, for the reasons already discussed. That is, if one failed to notice something when it happened, it is hardly surprising that he or she fails to remember it later. And yet the potential implications of this very particular kind of memory failure are impressive. Recent research has shown that change blindness can lead people to falsely identify perpetrators of crime (Davies & Hine, 2007; Nelson, Laney, Bowman Fowler, Knowles, & Loftus, 2011).

For example, Nelson et al. (2011) showed subjects a video depicting a student stealing money from another student. After the thief takes the money and tucks it into a book, she is seen leaving the room and walking down the hall. For half of subjects, the thief is then switched for another actor as she walks around a corner. Subjects in the change conditions were quite unlikely to notice this change of actors (just 4.5% noticed), but it certainly did affect their later memory for the video. The subjects who experienced the change of actors were as likely to identify the innocent actor (35%) as the actual thief (36%) in a subsequent lineup. By comparison, just 9% of subjects in the no-change conditions identified the innocent actor and 64% correctly identified the thief.

Biased Encoding

Because it is not possible (or desirable) to remember every detail about the world around us, we are left in a position where we are forced to make assumptions and inferences about that world. These assumptions save time and energy—think how inconvenient it would be if you could not make and rely upon assumptions about how, for example, cars work. When getting into a strange car you will likely need to adjust the seat and note whether there is a clutch, but you can safely assume that there will be a steering wheel, a gas pedal, and a review mirror (to say nothing of four wheels and an engine). But these assumptions also come at a cost. Because we tend not to notice aspects of events that fit with our expectations, we tend not to fully remember these details later (though we may think we do). Likewise, aspects of events that do *not* fit with our expectations, unless they earn special attention at the time of encoding, are unlikely to be remembered later. Note that biased encoding can lead to errors both of omission and of commission. Unnoticed details are not remembered accurately, while assumptions take over and fill in some details, and not necessarily correctly.

In an interesting demonstration of the importance of both attention and expectations, Friedman (1979) recorded subjects' eye movements as they carefully studied complex line drawings. The time that subjects spent looking at each detail of the drawings was determined by the appropriateness of each detail to the scene. So subjects spent approximately 4 times as long looking at details that did *not* fit in a scene (like a steamer trunk in an office scene) relative to those that did (like a chair). These attentional biases then led to biased memories. Details were changed between successive viewings of the pictures. Sometimes plausible details (like a sheep in a farm scene) were changed for other plausible details (like a goat). These changes tended not to be noticed or remembered. But other changes were less plausible (like a cow being replaced by a hippopotamus), and these were almost always noticed and remembered.

Schemata

Researchers at least as far back as Bartlett (1932; see also Alba & Hasher, 1983) have noted that our knowledge and beliefs about the world influence what and how we perceive and thus remember. Many parts of human life are predictable, based on past experience. Our memory systems take advantage of this predictability, by forming and utilizing schemata. Schemata are memory templates, describing the typical elements of the events (and places, information, etc.) that we repeatedly experience. Because our lives are full of redundancies, this prior knowledge is, in fact, quite helpful in predicting what will happen in the future. Because I have been in hotel rooms before, I know to expect a bed, a bathroom, a Gideon's Bible, and probably a TV, and also not to expect an ice cream maker. This is because my hotel room schema tells me that the former are common and the latter is highly unlikely. Because of a hotel room schema, the average adult upon entering a hotel room will toss his or her suitcase on the bed without first stopping to comment on the presence of a bed or bathroom or TV. By contrast, a small child who does not yet have a hotel room schema (because of lack of experience across multiple hotel rooms) may well be excited about any of these things.

Brewer and Treyens (1981) tested this type of schematic knowledge experimentally. They asked each of 86 undergraduate subjects to wait in a graduate student office on the pretence of the experimenter needing to check whether the study was ready to run. After a delay of 35 seconds, the experimenter returned, escorted the subject to a different room, and informed the subject that the true purpose of the study was to test subjects' memory for the first room. Subjects' memories were guided by what might be called their academic office schemata. For example, although they were highly likely to remember schema-relevant items like a desk and shelves, they were unlikely to remember items that were neither salient in themselves nor relevant to the office schema, like a picnic basket. But they were also likely to *falsely* remember seeing items that were highly relevant to the schema, like books, even when they had not actually been present in the room. Schemata amount to very convenient memory shortcuts, and our memory systems tend to rely on them. This reliance can lead to incomplete memories, because we do not notice and thus do not encode elements of our world that fit in with our schemata. For more on this problem, see the section on "Memory Reconstruction." Reliance on schemata can also lead to the blurring of similar and recurring events in our memories, and it can lead us to forget to perform repeated tasks, like taking medications (Schacter, 1999).

Motivation to Remember

People who are motivated to remember something at the time the memory is encoded sometimes show better memory than those who are less motivated (Loftus & Wickens, 1970). Loftus and Wickens gave subjects a paired-associates memory test, where different words were rewarded to differing degrees at test. Subjects were more likely to remember high-value words than low-value words, especially when they were told the values of the words as they were learning them, rather than after the fact. This differential impact of motivation at encoding versus retrieval has been supported by other work (e.g., Naveh-Benjamin, Craik, Gavrilescu, & Anderson, 2000). This result is apparently counterintuitive, however, as outside judges assume that all motivation to remember is equivalent, no matter the timing (Kassam, Gilbert, Swencionis, & Wilson, 2009).

Emotionality of to-be-remembered events is another key factor considered in other chapters in this book. To avoid redundancy, just one type of emotional response will be considered, briefly, here: stress. Unlike emotional arousal and even trauma, which normally improve the accuracy and fluency of memory (Conway et al., 1994; McNally, 2003; Peace & Porter, 2004; Shobe & Kihlstrom, 1997), stress has been shown to have a generally negative effect on memory (e.g., Morgan et al., 2004; see Reisberg & Heuer, 2007, for a review). Morgan et al. (2004) found that soldiers in an extreme wilderness and mock captivity training camp who experienced 40 minutes of extremely stressful interrogation were subsequently less able to identify their interrogators than other soldiers who experienced less stressful interrogation (see also Lieberman et al., 2005; Southwick, Morgan, Nicolauou, & Charney, 1997). That is, these highly stressed subjects had poor memories for the details of the interrogation that they had experienced, even though their interrogators had been demanding direct eye contact.

Even comparatively mild acute stressors like being asked to give a short speech can have a negative effect on memory, as can drugs that work to mimic these stressors (Payne, Nadel, Britton, & Jacobs, 2004). In a meta-analysis of studies of stress and memory, Deffenbacher, Bornstein, Penrod, and McGorty (2004) found that stress was a reliable impediment to accurate memory.

Stress, unlike other types of emotionality, seems to produce a defensive response that leads people to focus their attention on something, anything, apart from the source of the stress (Reisberg & Heuer, 2007). In this sense, it is like other sources of memory error discussed earlier. Stress alters attention and thus what is encoded and subsequently available for retrieval.

Effort and Goals During Retrieval

Memory retrieval is not a simple event—something that simply happens because we want to remember X—but rather a set of processes that are used in different ways depending on our particular goals. This is evident in the fact that sometimes it is worth searching hard in memory to try to recall exactly what happened and how, whereas at other times a vague notion of some element of the past suffices (see, e.g., Dalgleish et al., 2007; Davis & Friedman, 2007). In the former cases, relative to the latter cases, searching through memory can take time and effort (see Hockley, 2008). In addition to tradeoffs between completeness and speed, there are tradeoffs between completeness and accuracy. Specifically, in cases where we are only willing to accept details that we are sure are accurate, our memories are likely to be incomplete.

Retrieval cues are the specific reminders that link us back to specific memories (see Rajaram & Barber, 2008). These can come from the content of questions or from images, words, feelings, noises, or smells in the environment. One memory can also form a powerful retrieval cue for another memory. Some memories are easier to retrieve when one can match the circumstances in which the memory was encoded, a phenomenon called context-dependent memory (Eich, 1980). This explains why going back to one's childhood hometown or school as an adult can bring up memories that one has not thought of for many years.

Retrieval Failure

Memory retrieval happens smoothly and effortlessly most of the time, yet we are often painfully aware of the occasional breakdowns in this process, called retrieval failures. Nearly every time we speak or write or recall a personal anecdote, we do so fluently. Yet failures to recall a specific word or event can sometimes seem to be constantly occurring. (Whenever I tell someone that I study memory, I am immediately regaled with tales of this type of memory failure.)

One highly familiar type of retrieval failure is a so-called tip-of-the-tongue (TOT) state, where one experiences a failure to recall a word (often a name) but has a strong subjective feeling of it being stored in memory but just out of reach (see Brown, 1991; Shafto, Burke, Stamatakis, Tam, & Tyler, 2007). The phenomenon seems to be universal, but it is particularly common in older adults and bilinguals. Because less frequently used words are more likely to lead to TOT states, research suggests that TOTs are likely

caused by a breakdown of communication between semantic and phonological processes in the brain. That is, we know what we want to say, but we cannot quite remember how to go about saying it.

There is also evidence that the act of working to retrieve a particular piece of information can inhibit retrieval of other, closely related pieces of information (Anderson, Bjork, & Bjork, 1994; MacLeod, Saunders, & Chalmers, 2010). So-called retrieval-induced forgetting is normally demonstrated in studies where subjects learn lists of semantically related words, then are given practice at retrieving some (but not all) words from some (but not all) categories. Later, subjects are asked to recall the full lists of words. Typical results show improved memory for rehearsed words but also, less intuitively, poorer memory for unrehearsed words from categories where some words were rehearsed, relative to words from categories that were not rehearsed at all. Other research has shown similar effects with more complex materials (e.g., Koutstaal, Schacter, Johnson, & Galluccio, 1999; Shaw, Bjork, & Handal, 1995).

Memory Reconstruction

p. 237 Memory is not just accessed from storage; it is in many situations reconstructed at the point of retrieval (Barlett, 1932; see Davis & Loftus, 2007). ↵ That is, bits of the original event are combined with information learned before or since, as well as current thoughts and feelings, to produce a new, conglomerated memory. Note that at this stage we have moved from primarily discussing errors of omission to primarily discussing errors of commission.

Source Monitoring

Sometimes we remember information but fail to remember, or incorrectly remember, the source of this information. Daniel Schacter (1999, 2001) calls this the memory sin of “misattribution.” The source monitoring framework (Johnson, Hastroudi, & Lindsay, 1993; Lindsay, 2008; Mitchell & Johnson, 2009) explains that although we often lack specific information about the source of a particular memory (because memories do not come with attached tags labeling them as real or imagined events, for example, or with other specific source information), we can and do make assumptions about the source of a memory based on certain qualitative and quantitative characteristics of that memory. So a remembered dream feels subjectively different than a remembered experience, and some information (e.g., a personal anecdote from one’s childhood) is more likely to have come from some sources (one’s mother) than others (the nightly news). These subjective characteristics and knowledge-based assumptions work well enough for accurate source monitoring most of the time, but, as with other memory systems, there are opportunities for error.

Misinformation

The fact that information from multiple sources (including misidentified sources) can be blended to form reconstructed memories leads to the so-called misinformation effect (see Davis & Loftus, 2007). Misinformation is incorrect information that can be blended in with memory after the fact. The misinformation can be incorporated quickly after the original event, or much later, but the key is that this alteration of memory may occur without the knowledge of the rememberer, such that he or she may be entirely sure about a memory that is, in fact, quite compromised.

In typical studies of the misinformation effect, mock witnesses watch an event, often a crime, usually presented via slides or video (e.g., Loftus, Miller, & Burns, 1978). Later they are given extra information that is relevant to the event, often in the form of misleading questions or written summaries. Critically, this extra information is misinformation, and this misinformation leads to memory errors. (There is some debate in the field regarding whether the misinformation completely alters subjects’ memories for the original event, or, for example, creates a separate alternate version of the memory that leads to confusion; see Ayers & Reder, 1998; Brainerd & Reyna, 2005.)

In an early demonstration of the power of leading questions to cause memory errors, Loftus (1975) showed subjects a film of a car accident and then later asked a series of questions about the crash. The critical question varied between groups. One group was asked, “Did you see a broken headlight?” while another group was asked, “Did you see the broken headlight?” The latter group was significantly more likely to answer in the affirmative, even though the two questions varied by only a single (and phonetically similar)

word. The reason for that is that the second question is essentially informing the witness that there was a broken headlight, and asking whether he or she managed to notice it. The first question does not carry any presumption about the existence of a broken headlight. The similarity of the two questions means that there need not be any malice (or forensic context) for leading questions to distort memories—a simple slip of the tongue or mishearing of the question could suffice.

Recent research has sought to quantify the misinformation effect in new ways. Okado and Stark (2005) showed subjects events and then gave them misinformation about those events, while they were in a functional magnetic resonance imaging (fMRI) machine. Subjects were then given memory tests outside of the fMRI. The levels of activation in certain structures of the brain (notably the medial temporal lobe and prefrontal cortex) were different depending on whether subjects went on to report correct or erroneous information at test. That is, differences in brain activation during encoding predicted correct versus incorrect memory later.

The misinformation effect also happens outside of the laboratory (see Loftus & Castelle, 2000) and for memories that have nothing to do with crime (e.g., Wilson & French, 2008).

Social Influences

In the last few years, one particular type of misinformation has come to the foreground: social misinformation. Research has shown that misinformation from co-witnesses to events has particular power with subjects. Although the effects of co-witness comments on memory have been studied for decades (e.g., Loftus, 1979; Loftus & Greene, 1980), there has been a recent explosion in this research area (e.g., French, Sutherland, & Garry, 2006; Gabbert, Memon, & Allan, 2003; Gabbert, Memon, Allan, & Wright, 2004; Hope, Ost, Gabbert, Healey, & Lenton, 2008; Takarangi, Parker, & Garry, 2006; Wright, Mathews, & Skagerberg, 2005).

Witnesses often assume that other witnesses have seen and heard the same events in the same way as themselves, and thus trust these other witnesses' memories as much as their own—especially where their own memory is incomplete. As such, conversations between co-witnesses (which do happen outside of the laboratory; see Paterson & Kemp, 2006) can be detrimental to the truth, by leading people to “remember” far more than they actually encoded.

A new technique for studying social misinformation has recently been fruitful. Called the “manipulation of overlapping rivalrous images by polarizing filters” (MORI) paradigm (Garry, French, Kinzett, & Mori, 2008; Mori, 2003), it allows for two subjects to sit in front of a single screen to watch a video. The two subjects assume (quite reasonably) that they are seeing the same images, but because each subject is wearing a different type of polarized glasses, they are in fact watching different videos (projected onto the same screen, much like for 3D movies). Because the two subjects believe they have seen exactly the same event, they are particularly likely to allow their co-witness's memories of the video to affect their own.

There is also new evidence that what witnesses say to other witnesses can also bias their own memories for the witnessed events (Kopietz, Echtermoff, Niemeier, Hellman, & Memon, 2009). That is, not only does hearing about other people's memories lead to bias but so does telling other people about one's own. Specifically, Kopietz et al. had subjects tell a fellow student about a bar brawl they had seen on video. Subjects were told either that their audience liked or disliked one of the characters involved in the fight. Subjects adjusted their telling of the story based on this information (though only if their audience was assumed to be sufficiently like themselves) and subsequently showed biased memory for the original video, in the direction of their own retelling.

False Memory

The reconstructive nature of memory can have some extreme consequences. Sometimes memory is not merely distorted by suggestion but created out of whole cloth by suggestion. False memories can be created in laboratories using a variety of techniques (see Loftus & Bernstein, 2005), and they can also be a by-product of everyday memory reconstruction (e.g., Clancy, 2005; Sheen, Kemp, & Rubin, 2001; Taylor, 1965)—perhaps particularly in the context of techniques used by some therapists (see Loftus 1993, 2004; Loftus & Ketcham, 1994).

Suggestive techniques that have been used to produce false memories range from word list memorization tasks (e.g., Read, 1996; Roediger & McDermott, 1996) to computer-generated false feedback (e.g., Bernstein, Laney, Morris, & Loftus, 2005; Laney & Loftus, 2008) to intensive interviews and mock dream interpretation methods (Loftus & Pickrell, 1995; Mazzoni, Lombardo, Malvagia, & Loftus, 1999) to digitally altered images and hypnosis (e.g., Scoboria, Mazzoni, Kirsch, & Milling, 2002; Wade, Garry, Read, & Lindsay, 2002). But the reasons that the memory system falls for these tactics are not really different than the reasons for other types of errors already discussed. That is, these suggestions are believable (and thus lead to false memories) because one or more factors apply: They fill a void in our existing memories (i.e., we do not specifically remember the opposite to be true), they fit in with our expectations and are thus plausible, they link in with our existing knowledge, they come from reliable (and even authoritative) sources, and/or people are somehow motivated to believe them. (Note that, as with other types of errors, these factors need not all apply, but more and stronger factors can lead to more and larger errors.)

For example, Laney and Loftus (2008) used a false feedback manipulation to give undergraduate subjects false memories for three different emotional childhood events: being hospitalized overnight, catching their parents having sex, and witnessing a violent fight between their parents. In the false feedback procedure, subjects are asked to fill out a set of questionnaires on a particular topic—in this case emotional childhood events. They are then told (falsely) that their data will be entered into a special computer program that will provide specific feedback for them. This special computer program thus serves as the reliable and authoritative source of information in these studies. After a delay, subjects are given their “feedback” (which is, in fact, not specific at all but is the false memory manipulation) and then asked to fill out more questionnaires. Subjects frequently become more confident that they had experienced a particular event that had been suggested by the false feedback. They may also produce very specific detailed memory descriptions that conform to the feedback suggestions.

p. 239 In this study, some 21% of manipulated subjects ended up forming false memories for one of the three false events (this ranged from 15% for the parent fight item to 38% for the hospitalized item). In other studies using the false feedback procedure with less controversial items, this proportion has ranged as high as 53% (Laney, Morris, Bernstein, Wakefield, & Loftus, 2008).

Because the suggested events in the Laney and Loftus (2008) study were at least a decade in the past for subjects, many did not have specific memories to contradict them. For some (but not all) subjects, suggestions that they had been hospitalized or even witnessed parental violence were plausible because they fit in with their general understanding of their own childhoods, even when they did not remember these events happening to them (cf. Lampinen, Meier, Arnal, & Leding, 2005). For other subjects, these events were less plausible or fit less well with their other memories, and were thus more likely to be rejected. (For further discussion of plausibility in the formation of false memories, see Hart & Schooler, 2006; Mazzoni, Loftus, & Kirsch, 2001.) These subjects were probably not particularly motivated to believe any of the three false events in this study, but this merely suggests that there is scope for even more people to believe (and believe more strongly) in contexts where they are motivated to believe (see Sharman & Calacouris, 2010).

Once established, false memories can be extremely compelling for those who have them (see Loftus & Bernstein, 2005). This leads to great difficulties in telling established false memories apart from true memories. This is particularly problematic in court cases where one party claims to remember a horrible event (like child sexual abuse), while another party claims that no such event occurred. Researchers have identified numerous possible characteristics on which true and false memories might theoretically be distinguished—from quantity of detail and confidence to consequentiality and emotional content—but as yet no definitive factor has been found that categorically discriminates between true and false memories. That is, false memories, like true memories, can be confidently held, detailed, long lasting, consequential, and even emotional, such that none of these characteristics can be used to guarantee a particular memory’s truth (Laney & Loftus, 2008; Laney, Bowman Fowler, Nelson, Bernstein, & Loftus, 2008; Laney, Morris et al., 2008; Loftus & Bernstein, 2005). As such, when memory errors are at their most extreme, they can also be very difficult to correct.

Conclusion

Because memory is not a static storehouse of information, but rather a set of keenly adapted dynamic processes, it is infinitely more efficient for our everyday needs. This efficiency has a downside, though, in the form of proneness to various types of errors, including errors of omission and of commission. These errors show broad commonalities, in that they tend to result from the combination of incomplete original information and inferences (based on experience or assumptions), plus perhaps confusion about the sources of different bits of information.

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